

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Supremo Green Revolution (SGR)

Submitted by: EcoPump Africa Ltd. (EPA)

Date: October 8, 2025

1. Executive Summary

EcoPump Africa Ltd. (EPA) is pleased to submit this proposal in response to the SGR RFQ for solar-powered irrigation systems in Kenya, Ghana, and Tanzania.

EPA is a regional engineering and agricultural technology company operating since 2014, with completed irrigation and solar-water-supply projects in Malawi, Zambia, Uganda, and western Kenya. Our company specializes in hybrid solar pumping solutions designed for smallholder farmers seeking reliable, affordable irrigation options.

EPA's solution offers a strong balance of technical capability, affordability, and practical field experience. While not as digitally advanced as larger multinational competitors, EPA provides robust on-site implementation capacity, strong local staff presence, and a proven farmer training framework. Our proposal aligns well with SGR's objectives under the Green Resilience Initiative.

2. Company Profile

EcoPump Africa Ltd. is headquartered in Lusaka, Zambia with regional offices in Kisumu (Kenya) and Jinja (Uganda). The company employs 46 full-time staff, including 8 mechanical engineers, 5 solar technicians, and 6 agronomy specialists.

EPA maintains the following certifications:

- ISO 9001:2015 – Quality Management Systems
 - ISO 14001:2015 – Environmental Management Systems
- (ISO 45001 certification is scheduled for 2026.)

EPA's past clients include the Malawi Irrigation Board, the Uganda Smallholder Support Agency, and several EU-funded rural agriculture programs. The company has installed over 240 irrigation systems since 2016, mostly between 1 and 4 hectares in size.

3. Technical Approach

EPA proposes a hybrid solar-pumping model tailored to shallow boreholes and smallholder drip irrigation.

Core System Components

- Solar PV arrays (2–4 kW, Tier 2 monocrystalline modules).
- DC or AC submersible pump depending on water depth (6,000–12,000 L/day).
- Reinforced 5,000 L polyethylene storage tanks.
- UV-resistant drip irrigation kits for vegetables and legumes.
- Optional 7-inch tablet dashboard for recording daily water use and pump runtime.

Key Design Principles

- Prioritize durability and maintenance simplicity.
- Minimize moving parts through direct-drive systems.
- Provide clear documentation for local technicians.
- Ensure systems can be expanded with add-on panel kits.

EPA does not currently offer GSM-based real-time monitoring but uses a rugged tablet-based “FieldTrack” data entry app operated by local extension officers.

4. Deliverables

1. Ten (10) hybrid solar irrigation installations (approx. 0.5–3 ha each).
2. Training for 480 farmers across all three countries.
3. Local technician training (20 technicians total).
4. Bilingual manuals (English, Swahili; Ghanaian languages optional).
5. Semi-digital monitoring using FieldTrack (tablet-based).
6. Eighteen-month warranty on pumps and controllers; 10-year solar panel warranty.

5. Compliance and Quality Standards

EPA complies with:

- ISO 9001:2015
- ISO 14001:2015
- IEC 60364 (electrical installation guidelines)
- National labor, safety, and installation codes

EPA acknowledges that ISO 45001 compliance is pending and that digital logging does not yet fully meet IEC 62257 recommendations for remote monitoring.

6. Implementation Plan

Phase	Activity	Duration
1	Field Assessments & Surveys	3 weeks
2	System Design & Procurement	4 weeks
3	Installation & Commissioning	5 weeks
4	Training & Handover	2 weeks
5	Monitoring & Support	Ongoing
Total estimated duration:		14 weeks.

7. Financial Proposal (USD)

Item	Quantity	Unit Cost	Subtotal	Solar Irrigation Systems	10	\$6,300	\$63,000	Farmer Training Sessions	10	\$1,900	\$19,000	Technician Certification	20	\$150	\$3,000	Spare Parts & Kits	10	\$450	\$4,500	Monitoring Tablets	5	\$600	\$3,000	Logistics & Transport	1	\$9,800	\$9,800	Total Estimated Cost	\$102,300
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Notes:

- Pricing excludes VAT.
- Warranty period (18 months) below SGR’s requirement.
- Valid 90 days.

8. Sustainability and Community Integration

EPA emphasizes long-term sustainability through:

- Spare parts depots in Kisumu and Jinja.
- Inclusion of 40% women participants in training.

- Annual refresher workshops.
- Cooperative■managed replacement funds.

Limitations include the absence of GSM monitoring and limited centralized support capacity.

9. Strengths and Weaknesses

Strengths

- Strong local presence and field-tested systems.
- Affordable and maintainable hardware.
- Gender■inclusive training and cooperative support models.
- Tablet■based data logging improves basic performance tracking.

Weaknesses

- Not ISO 45001 certified.
- No real■time monitoring solution.
- Warranty below required duration.
- Equipment less advanced than Tier■1 competitors.

Overall, EPA represents a reliable and practical mid■tier option with room for improvement.

10. Conclusion

EcoPump Africa Ltd. appreciates the opportunity to participate in this strategic initiative under SGR's Green Resilience Program. EPA provides a practical, reliable, and community■centred approach capable of meeting core technical, social, and economic objectives.

Submitted by:

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